SEMESTER-VII

URA701: AUTONOMOUS SYSTEMS AND MACHINE VISION				
	L	T	P	Cr
	2	0	2	3.0

Course Objective: To become familiar with the concept of robotics and learn different types of algorithms for process uncertainties, data fusion and localization.

Syllabus

Fundamentals of Robotics & Automation: Intelligent Robots, Control Systems and Components, Robot Motion Analysis and Control, Robot End Effectors, tactile and vision sensors in robotics, Path planning in joint and task space, Obstacle avoidance and optimal planning, Review of robot control methods, Optimization in robotics, Human-robot interaction, Manipulation, Tendon driven manipulator.
Sensors and sensing: Model of Sensors & Process uncertainties, Introduction to estimation, estimation methods & relation between different estimators, State space modeling, LTI Systems & Kalman Filter & Extended Kalman Filter, Other Navigation Filters including Bayesian Filters, Information Filters, Particle Filter etc.
Sensors used in Robotics: Accelerometer, Gyro, Compass, Encoder, Laser, Ultrasonic Sensor, Camera, Sonar, InfraRed Sensor, Tactile Sensor etc. Multisensor Data Fusion Fundamentals; INS, GPS Aided Navigation & Data Fusion, Future Trends, Applications, Feedback Control of Dynamic Systems.
Machine Vision: Image Transformations, Camera Projections, Camera Calibrations, Depth from Stereo, 3-D Reconstructions, Segmentation, 2D and 3D LiDAR, Introduction to multi-sensor calibration.
Computer Vision & Robotics: Video Frames, Key frames, Background Subtraction, Moving Object Detection, Robot Localization, SLAM.

Laboratory Work
To implement various techniques studied during the course.
1. Introduction to Joints, DH-Parameters, and Applications of Autonomous Systems in MATLAB.
2. Introduction to Inverse Kinematics, Forward Kinematics in MATLAB
3. Consider a 2-DOF serial robot [both rotatory joints], calculate the DH parameters, and perform the Inverse kinematics with the graphical representation of the serial robot in MATLAB.
4. Introduction to the Robo-Analyser [Student access is free] software.
5. Camera Calibration
6. Stereo and Tensor Calibration
7. Estimation of the Pose of the Object with respect to the Camera and World coordinates systems.
8. Implementation of Kalman Estimator.
9. Estimation of the Pose of the Object (using feedback control) with respect to the Camera and World coordinates systems.

Course Learning Objectives (CLO)
The students will be able to:

1. Analyze the different robot components and their control systems.
2. Analyze path planning and Robots Motion.
3. Design and Deployment of various sensors and their analysis.
4. Develop and Analyze uncertainty estimation and data fusion algorithms.
5. Design and Develop a human-robot interaction system.
6. Analyze and Synthesize vision-based robot localisation and path planning methods.

Text Books

1. Hands-On Robotics Programming with C++: Leverage Raspberry Pi 3 and C++ libraries to build intelligent robotics applications, Dinesh Tavasalkar, PACKT (2019)
2. Multi-Sensor Data Fusion with MATLAB, Jitendra R. Raol ,CRC Press (2010)

Reference Books

1. Mobile Robot Localization and Map Building, José A. Castellanos, Juan D. Tardós, Springer (2012)

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessional (Assignments/Projects/Tutorials/Quizzes)	35